WEB Based Automatic Classification of Musical Instrument Sounds

Mark Williams, Ian Kaminskyj
Faculty of Engineering, Monash University
email: markwil@yoyo.cc.monash.edu.au
email: ian.kaminskyj@eng.monash.edu.au

Abstract

A Web based automatic musical instrument classifier has been developed. It is intended primarily to assist researchers working on the problem of automatic identification of musical instrument sounds. It is a web-based implementation of the classification system developed by Ian Kaminskyj. The system operates on unknown sound files, detecting the presence or absence of vibrato and extracting five features. The system uses the features to identify the most likely instrument to have produced the test input. Java and Java 3D has been used to provide a system that is platform independent. This web based system produces graphical representations of the features extracted from the unknown recording. JAVA3D allows for rotation, zooming and translation of three dimensional feature waveforms. The WWW site provides interactivity, a graphical user interface and ease of use.

1 Introduction

There are many software tools available to researchers for processing sound files, e.g. Pitch Detection SDK (www.mb-usa.com/english/pdsdk.shtml) for automatically detecting and correcting pitch. There are a number of useful web sites for researchers working on the problem of analysis of musical instrument sounds. These include: The Studio Online (SOL) Project (www.ircam.fr/produits/technologies/sol/index-e.html), the Sandell Harmonic Archive (SHARC) Database (http://sparky.parmly.luc.edu/sharc/) and Tae Hong Parks' feature extraction software (http://silvertone.princeton.edu/~park/thesis/).

The SOL project not only provides a source for musical sounds but also furnishes an interactive feature that allows for online transformation (stretching, transposition, filtering etc.) of reference sounds together with the users own sounds. The McGill University Master Samples (MUMS) provides a number of reference CD's with musical instrument recordings. They have been utilised on the SHARC website, wherein static plots of predetermined spectral analysis results are carried out on a number of recordings. The feature extraction software described by Park (2000) is a Java application that provides useful tools including root mean square (RMS) amplitude envelope/spectral centroid calculation and pitch detection (currently, the software is not readily available on the WWW).

A World Wide Web site to automatically classify musical instruments and dynamically calculate features has been developed. It is intended that the site will further assist researchers in both the analysis and automatic recognition of musical instrument sounds.

2 Theory

The classifiers system structure is shown in Figure 1. The vibrato detector firstly determines the presence of any vibrato within the input test sample. If vibrato is deemed to be present, the vibrato instrument classifier is invoked, which classifies the input as having been produced by one of three possible vibrato instruments: cello, flute and violin. Alternatively, if no vibrato exists, the non-vibrato instrument classifier is executed. It classifies the input as having been produced by one of nineteen possible non-vibrato instruments.

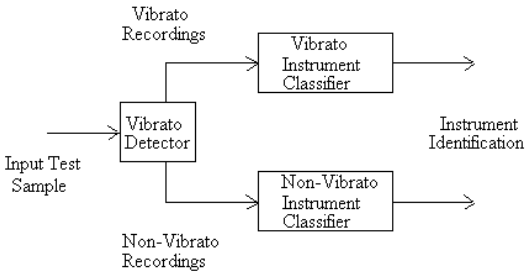


Figure 1 Classification System Structure

The vibrato/non-vibrato instrument classifier is shown in Figure 2 and uses five single feature classifiers to independently determine the most likely instrument to have produced the input test sample. The features extracted by the system and thereby used by the classifier include the Root Mean Squared (RMS) amplitude envelope, cepstral coefficients (CCC), Multidimensional Scaling Analysis trajectories (MSA), Constant Q Transform (CQT) frequency spectrum and the spectral centroid (SCD). Kaminskyj (2000) describes the features in detail. Five classifiers (vibrato or non-vibrato depending on the result of the detector) use these five features to independently identify the test sample. The results of the individual classifiers are combined together via a confusion matrix accuracy and reliability-weighting scheme which produces an estimate of the most likely instrument to have produced the test input. Kaminskyj (2002) provides further details regarding the classification scheme.

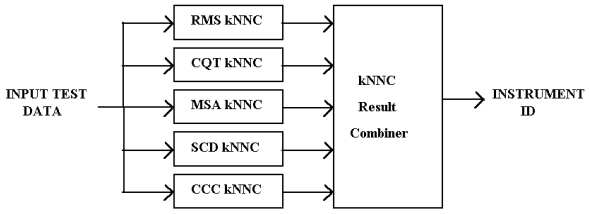


Figure 2 Classifier Details

3 Implementation Method

HTML, JavaScript and Java and were used to develop the web-based system. HTML coupled with JavaScript provides more functionality than solely using HTML. JavaScript was, for example, used to validate the user's input (parameters) before being passed to the musical instrument classifier program. Java provides many classes for dealing with file I/O and drawing objects and is a platform independent language. It can also be used to create either standalone applications or java applets that are executed over the web. Java's Abstract Window Toolkit (AWT) was chosen to implement the output feature plots rather than the more recently developed Java Swing classes. The rationale behind this was that the AWT classes provide all of the functionality required and do not require additional WWW browser plugins. Unlike Java Swing, it is compatible with older versions of Java. The integrated development environment used to write the software was Sun Microsystems' Forte for Java.

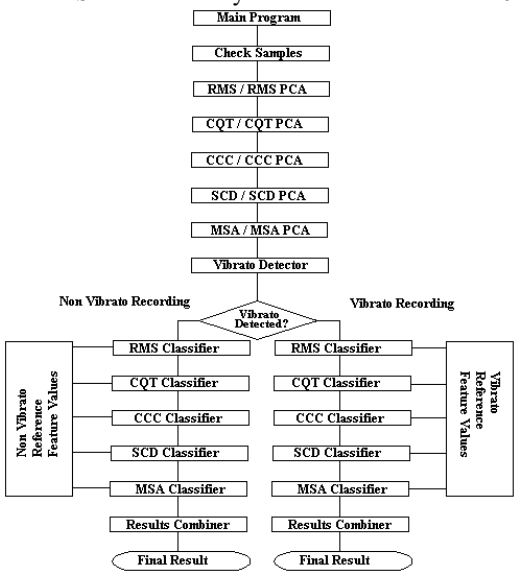


Figure 3 Web classifier software flow chart

Figure 3 shows a software flow diagram, illustrating the order in which the programs are executed and the way the system classifies the instruments.

The main controlling program is a web page form that accepts the various parameters (name of the WAV file to be uploaded, pitch information, which feature classifiers to bypass, classifier search limits etc) from the user and validates them.

The primary purpose of the main program is to execute the sundry programs in sequence and to pass user parameters. The sequence of the programs is significant as results of some programs are used as input to subsequent programs (for example the output of the feature extraction programs is used as input to the principal component analysis programs). After the user-defined parameters have been verified, the length of the input WAV file is examined to determine if there are enough samples to perform the feature extraction (if there are not enough samples the applet will stop and generate an error message). Assuming that there are enough samples, the header information of the WAV file is removed, leaving the

raw data in a text file format, suitable for use as the input to the feature extraction programs.

Five features are then extracted from the unknown WAV file directly. These feature waveforms are then used as input to the principal component analysis (PCA) programs, which reduces the amount of information processed by the classifier (Jollife 1986); PCA feature waveforms are thereby obtained.

The file is then checked for the presence of vibrato. The vibrato check is a fundamental feature of the system (as it reduces the number of reference recordings that the unknown recording needs to be compared to during classification). The main program runs a set of vibrato/non-vibrato classification and combiner programs (as illustrated in Figure 1) based on the vibrato detection result obtained. The parameters of the vibrato detector can be changed to allow for different thresholds of vibrato: the parameters comprising the minimum amplitude, and upper and lower frequency limits.

Following the vibrato detection, each feature (either direct or PCA) is used by an individual feature classifier to identify the input sound.

The RMS, CCC, CQT, MSA and SCD classifiers compare the direct or PCA features of the unknown input recording to a set of reference recordings and select the closest match via the *k* Nearest Neighbour Algorithm (*k*-NNC) (Duda & Hart 1976).

3 Graphical User Interface

The parameter settings are obtained by a HTML form, which has been broken into two parts. Figure 4 shows the first component of the input form; here the user specifies the unknown input WAV file to be uploaded together with its pitch. (The system requires that the pitch is known *a priori* and there are a number of programs available to perform this function (e.g. Pitch Detection SDK (www.mb-usa.com/english/pdsdk.shtml), TUNE!IT (www.zeta.org.au/~dvolkmer/tuneit.html) and RBC Voice Tweaker (www.thedirectxfiles.com/features/rbcvt/rbc1.htm)). The user can turn vibrato detection on/off, alter the threshold values of the vibrato detector and enable/disable any of the five individual feature classifiers. Figure 4 shows the default settings of the classifier that were determined to be optimum for the system. For vibrato recordings, CQT, SCD and RMS feature classifiers are excluded from classification. For non-vibrato recordings however, the RMS feature has been included. The user can modify these settings.

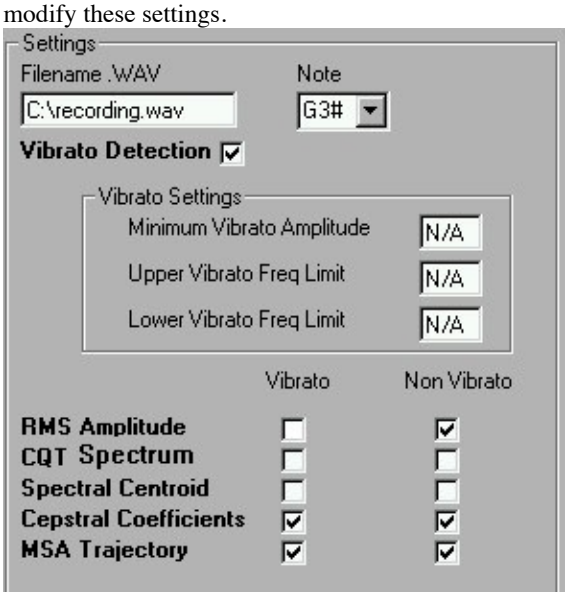


Figure 4 Input form

The second part of the input form is the classifier and combiner settings, shown in Figure 5. These settings allow the user to change the classifier and combiner settings to improve the accuracy of the result. The instrument elimination check box allows the classification program to exclude instruments when comparing the unknown input WAV file to the reference recordings. For example, when the unknown instrument sound has a pitch that is outside the playing range of an instrument, the classifier will exclude this instrument from classification (e.g. this would ensure that a marimba C3 test tone was never misclassified as a glockenspiel, an instrument whose note range does not extend below G5). The confusion matrix accuracy and reliability settings of the combiner can also be adjusted by selecting from the drop down box. By default, the combiner is based on an empirically determined reliability confusion matrix weighing scheme. The settings however, can be changed to have additional schemes such as (a) based on accuracy and reliability, (b)

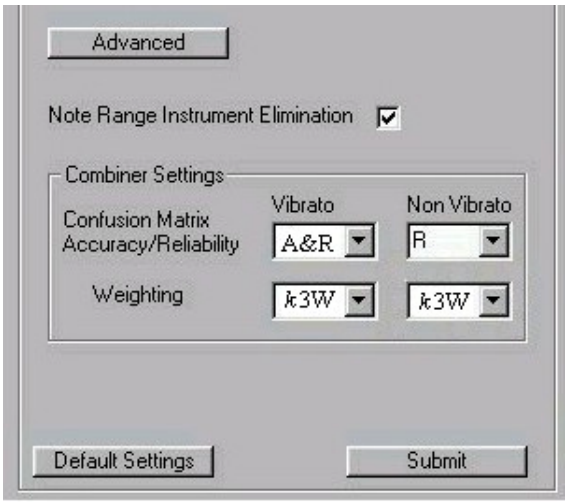


Figure 5 Classifier and Combiner parameters

solely based on accuracy only, or (c) a scheme independent of both accuracy and reliability.

The combiner NNC majority weighting scheme can also be adjusted to $k1$, $k3$ unweighted/weighted and $k5$ unweighted/weighted NNC majority weighting scheme. By default, the weighting for the combiner is set to the $k3$ weighted majority voting scheme. The individual feature classifier settings can be adjusted by pressing the advanced button, opening a window of options shown in Figure 6.

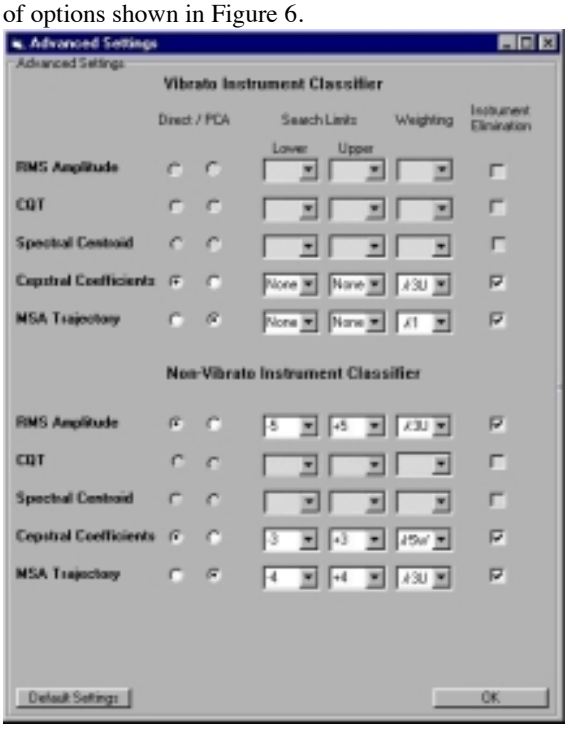


Figure 6 Advanced Settings

As can be seen the classifier settings can be changed to limit the classification to a smaller number of references occupying a restricted range of notes. The search limits indicate what note range to restrict the comparison of references with the test input. The direct or PCA features can be selected for use by the individual classifiers. Instrument elimination for each feature classifier can be individually enabled/disabled by selecting the appropriate checkbox. This however, is only possible if the instrument elimination checkbox on the main parameter page was checked. The weighting of each feature classifier can be set in the same way as the combiner settings of Figure 5. Pressing the OK button will submit the user chosen parameters to the classifier and return the user to the main page. Note that the options for RMS, CQT and SCD are not available under the vibrato recording settings because they were not been selected in Figure 4. Similarly, for the non-vibrato classifier, the CQT and SCD options were not available.

The graphical results of the feature calculations are displayed in the right hand side of the web page as illustrated in Figure 7. After pressing the submit button, the user is presented with plots of the enabled feature waveforms for the unknown recording. For comparison purposes, the corresponding waveforms of the closest reference that was found for each enabled individual feature classifier is also shown. Again note that the RMS, CQT and SCD feature plots and closest references are not shown because they were disabled for the vibrato instrument classifier in the settings of Figure 4 (the unknown sound file was found to contain vibrato).

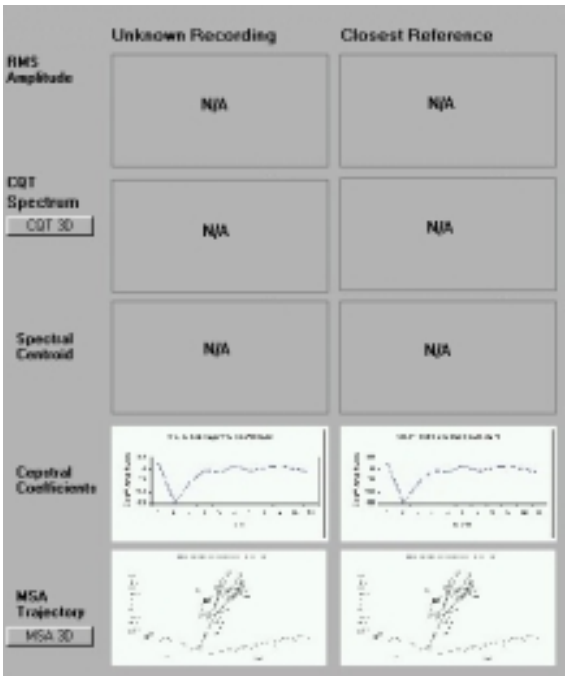


Figure 7 Graphical output of features

Finally, the results of the automatic classification system are displayed at the bottom of the page as shown in Figure 8. The vibrato detection result is displayed, together with the individual feature results. The overall result based on the combiner settings used is also presented.

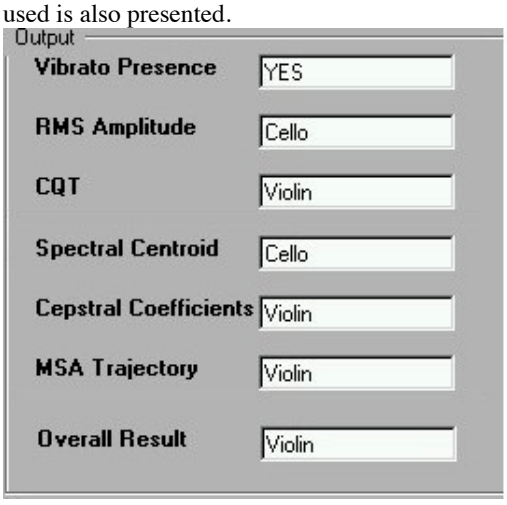


Figure 8 Results pane

4 An Example Session

In order to clarify how the WWW page would typically be used, a hypothetical example session will be detailed. In the sample web page shown in Figure 9, the user uploaded the unknown source file HORD3SC.WAV (a French horn playing the note D3 sharp). The pitch of the unknown input WAV file has been selected from the drop down box (pitches C3 to C6 are available for selection). The vibrato detection feature was enabled and its settings were not changed. Therefore the threshold default values were used (minimum vibrato amplitude = 20 cents, upper vibrato frequency limit = 8 Hz and lower vibrato frequency limit = 5 Hz). The user elected to include all feature classifiers by selecting all five checkboxes. If, for example, the user chose to deselect the RMS feature, the RMS feature classifier would not run and hence not be used in the combination of the individual feature classifier results.

Figure 10 shows the advanced settings for this session. As the user elected to use all features, the advanced feature settings were available for all individual feature classifiers. Only the direct feature values were used during classification and instrument elimination was enabled for all feature classifiers. A number of different search limits and weightings were used on each feature classifier. For example, the non-vibrato RMS feature classifier search limits were set to +/- six notes (an octave). This restricted the RMS classifier to six notes above the unknown input WAV file pitch and six notes below, when comparing it against the reference instruments. The user then set the combiner settings on Figure 9 to use an accuracy and reliability confusion matrix weighting scheme for both the vibrato and non-vibrato instrument classifiers. The accuracy and reliability confusion matrix weighting scheme uses empirical accuracy and reliability results to weight the results of the individual feature classifiers.

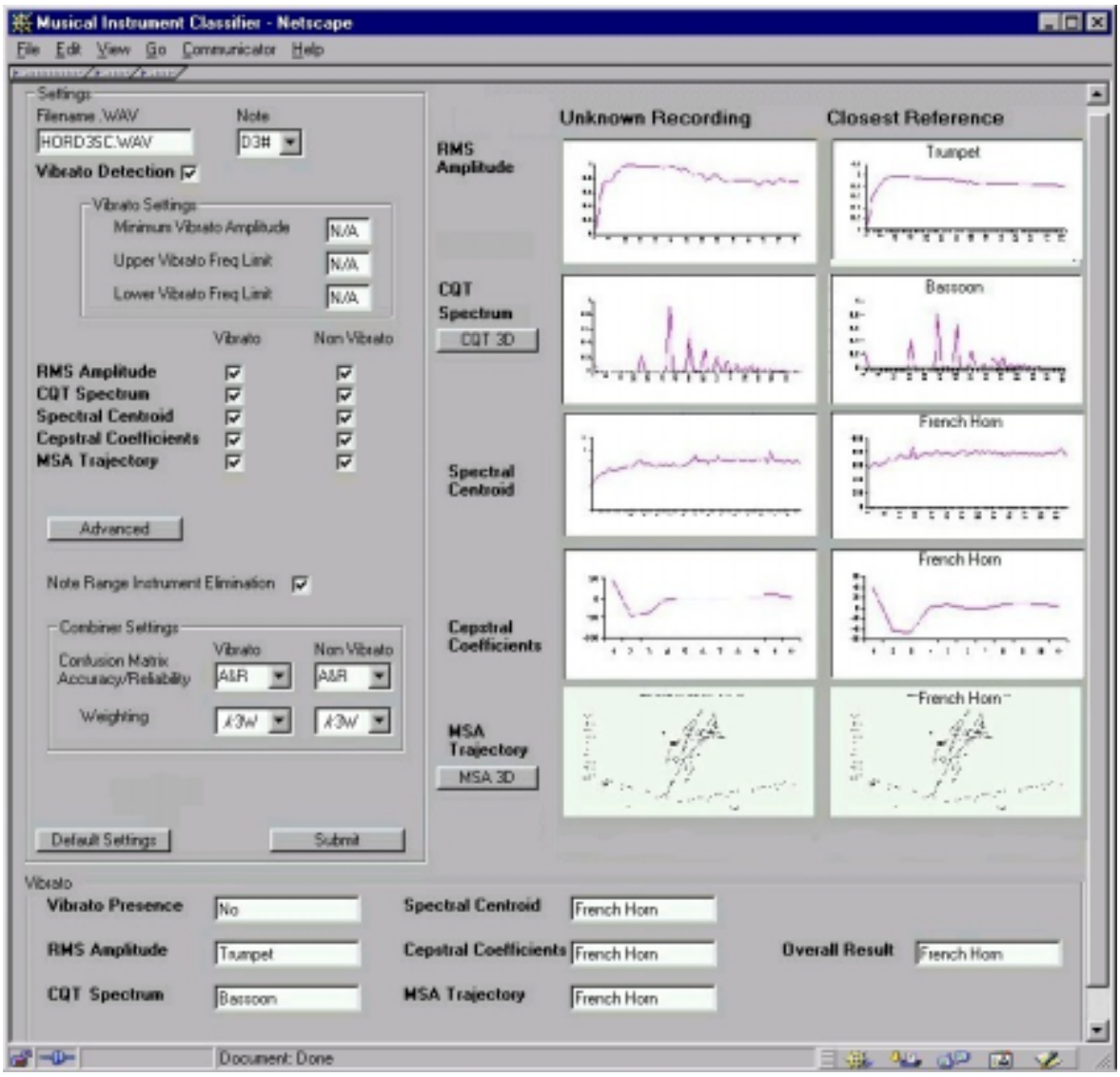


Figure 9 Sample Session

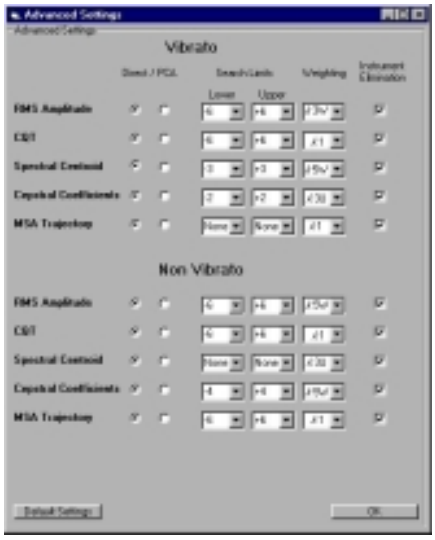


Figure 10 Advanced Settings: Sample Session

The submit button was used to start the program, after which the feature graphs and results appeared. (The default button restores the parameter settings on the form to their default values.)

The graphs plotted in Figure 9 show the RMS, CQT, SCD, CCC, and MSA extracted feature waveforms. To the right of these graphs is the feature data for the closest reference plotted. This allows for a visual comparison of the calculated feature and the closest reference found. In the Figure 9 example, the output on the right hand side shows an RMS Amplitude plot of a trumpet which was deemed by the RMS classifier to be the closest reference to the RMS amplitude envelope of the unknown input WAV file. Below this is a CQT plot of a bassoon, deemed to be the closest reference to the unknown input WAV file by the

CQT classifier. The CQT 3D button plots a 3D spectral waterfall plot of the unknown input recording as shown in Figure 11. The Java 3D graph allows the user to zoom in/out rotate and translate the graph. This feature is also provided to plot the MSA trajectory by pressing the MSA 3D button.

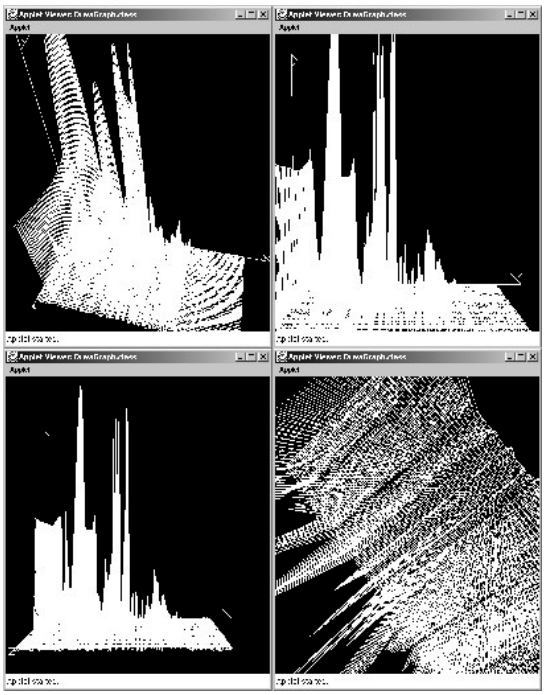


Figure 11 Java 3D: CQT Spectral Waterfall

The spectral centroid, cepstral coefficient and MSA trajectory plots of the closest reference are of a French horn, which is, as we know the same instrument as the input file. While three out of the five feature classifiers were correct, the combiner weighted these results based on the chosen accuracy and reliability weighting scheme. As can be seen at the bottom of Figure 9, the overall result was that the most likely instrument to have played the test input sound was a French horn.

4 Conclusion

The web based automatic musical instrument classifier presented provides remote access to sound processing and analysis via an interactive, platform independent application. The system can be readily distributed to the user via web applet (located at http://www-personal.monash.edu.au/~kaminski/) or as a standalone Java application that can be run offline. There is much scope for advancement including: supporting different sound file formats, providing automatic pitch detection, the ability to update the reference files to include test tones (which will enhance the general performance of the system Kaminskyj 2002), reference instrument elimination (i.e. if it is known beforehand that some instruments were absent from the test set recordings), expanding program parameter options, the ability to classify multiple recordings and extract new features to help identify instruments.

References

Kaminskyj, I., 2000, “Multi-feature Musical Instrument Sound Classifier”, *Proceedings ACMA Conference,* pp. 46-54.

Kaminskyj, I. 2002, "Multi-feature Musical Instrument Sound Classifier w/user determined generalisation performance", Proc. Aust. Comp. Music Assoc. Conf. 2002 Melbourne, Australia, July 2002, (accepted for publication)

McGill University Master Samples, http://www.music.mcgill.ca/resources/mums/html/mums.html [Accessed 27/04/02]

The Studio Online Project, www.ircam.fr/produits/technologies/sol/index-e.html [Accessed 27/04/02]

Sandell, G.J, 1995, “SHARC Timbre Database”, http://sparky.parmly.luc.edu/sharc/ [Accessed 27/04/2002]

Park T.H, 2000, "Salient Feature Extraction of Musical Signals", http://silvertone.princeton.edu/~park/thesis/ [Accessed 29/04/2002]

Jollife, I. T., “Principal Component Analysis”, Springer-Verlag, 1986.

Duda, R. O., and Hart, P. E., “Pattern Classification and Scene Analysis”, Wiley-Interscience, 1973

Pitch Detection SDK, 2001 www.mb-usa.com/english/pdsdk.shtml [Accessed 27/04/02]

TUNE!IT - Musical Instrument Tuning Software for Windows, 2001, www.zeta.org.au/~dvolkmer/tuneit.html [Accessed 27/04/02]

RBC Voice Tweaker, 2001, www.thedirectxfiles.com/features/rbcvt/rbc1.htm [Accessed 27/04/02]